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Dual Resource Constrained Scheduling for Quality Control Laboratories

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Abstract: This work presents a novel formulation for quality control laboratory scheduling considering both equipment and analysts as constraints. The problem is modelled as a dual-resource constrained flexible job shop problem. The formulation considers analyst tasks at different times during the processing of samples. The problem is formulated as a mixed integer linear programming model (MILP) aiming to minimise makespan. Two sets of instances for the scheduling problem are proposed. The first instance consists of a small example that illustrates the proposed formulation and is solved to optimality. The second instance mimics the real industrial problem and shows the challenges resulting from growing complexity.

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Keywords: Scheduling; dual resource constrained; quality control; flexible job shop; mixed integer linear programming.

1. INTRODUCTION

Recent trends are shifting industrial paradigms and enabling companies to cut on costs and save time on both services and production through better planning and human-machine interfaces, better production control, raw material availability control, inventory levels, and energy consumption, Rüßmann et al. (2015). With the right decision making in place, the relationship between customers and suppliers can be significantly improved and, with the increase in profit margins, companies have the flexibility to better fulfil their social and environmental responsibilities.

Although all industry sectors are affected by these changes, one where it is inevitable is the chemical sector. Chemistry 4.0, involving the chemical and pharmaceutical industries is experiencing changes with great potential impact, Falter et al. (2017). With the electric car, a decline in demand for many chemical products, such as gasoline-resistant plastics, oil and fuel additives, is expected. Similarly, in the pharmaceutical sector with the patent expiration and price controls increasing in Europe and the US, profit margins are under threat, Maslaton (2012). The prospect of these changes pressure companies to invest and try to find solutions, both in process improvement and product portfolio diversification.

It is within Industry 4.0 scope, and more specifically Chemistry 4.0, that the scheduling and rescheduling problem arises. Knowing the current state of operations, described by data on demand and plant state, it is possible to find the best possible set of decisions to optimise productivity, minimising costs and/or maximise revenues.

Quality control (QC) laboratories are one of the key structures in many chemical process industries. Because they are upstream and downstream operations (suppliers to manufacture and close to the point-of-sale), a loss in efficiency may compromise production start or delay shipments, Maslaton (2012). Therefore, QC laboratories have been appointed as an area where scheduling could have a very high impact on efficiency, Schäfer (2004).

This work focuses on QC laboratory scheduling considering both equipment and analysts as constraints. To do so, a mathematical formulation of a dual-resource constrained flexible job shop problem is proposed, where assignment of jobs to equipment and analysts is performed in order to minimise maximum completion time. The peculiarity of the model is that it considers analysts' tasks, not only in mounting and dismounting, but also at specific times during processing. The mathematical model is implemented as a mixed integer linear programming model (MILP). To the best of our knowledge the mathematical formulation developed and application to the QC laboratory scheduling problem has not been presented previously. Additionally, two sets of instances are proposed. The first is an illustrative example that is solved to optimality. The second simulates the real QC laboratory scheduling problem.

The remainder of this paper is organised as follows. The rest of this section presents a review on related work. Section 2 describes the QC laboratory scheduling problem. Section 3 gives a detailed description of the developed mathematical model. Section 4 presents the developed instances and results obtained. Finally, in Section 5, con-

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cluding remarks are made and directions for future work are presented.

1.1 Related work

In terms of dual-resource constrained scheduling, Scholz-Reiter et al. (2009) did a comparison for static dual-resource constrained scheduling between MILP and various dispatching rules. Dispatching rules were also compared using long-term simulation. The application use case was a semiconductor manufacturing system. However, workers only had tasks at the beginning of task processing, for loading, and at the end, for unloading. Lei and Guo (2014) presented a variable neighbourhood search (VNS) algorithm for a flexible job shop scheduling with workers. Their algorithm was tested against extensions of classic flexible job shop instances to include workers. Although the presented VNS yielded good results, they assume a worker has to be assigned to the task during the whole machine processing time. Ciro et al. (2015) studied an open shop scheduling with worker flexibility. They implemented a MILP model and compared it with a proposed ant colony algorithm tuned with fuzzy logic controllers. In the implemented scenario, workers were assigned to jobs for the full length of their processing time on the machine. Furthermore, because their application is an open shop problem, jobs have just one operation to perform.

In terms of quality control specific applications, Ruiz-Torres et al. (2012) addressed the QC laboratories scheduling of technicians by minimizing flow-time, using a heuristic consisting of a combination of dispatching rules. In Ruiz-Torres et al. (2017) the assignment of work to technicians is dependent on preferences and train related certification. In Costigliola et al. (2017) and Lopes et al. (2018) simulation models of QC laboratories were developed. Costigliola et al. (2017) developed a simulation model for the analytical workflow in the laboratory. Through the analysis of different simulation scenarios, Lopes et al. (2018) studied the impact of different governance strategies in process flow efficiency in the laboratories.

In this paper, a dual resource constrained scheduling problem formulation is proposed to model the scheduling problem in QC laboratories as a MILP. The formulation considers cases where analysts/workers are not only required for setup and teardown activities, but also for tasks at specific times during processing operations.

2. PROBLEM DESCRIPTION

Quality control receives samples for analyses, depending on the plant type, before, during and after manufacturing to certify product characteristics, Ruiz-Torres et al. (2012). Attaining the specific characteristics of a product requires various analytical procedures. When a sample arrives at the laboratory it may need to be analysed through one or more techniques, involving, or not, precedence between them, Schäfer (2004).

A sample analysis requires not only the specific equipment to perform the analysis but various interventions from an analyst. The analyst is usually needed to prepare the sample, setup the equipment, run the test and dismount the equipment, Lopes et al. (2018). These tasks may be required at specific times during the analysis or may take the whole time an analysis is run.

In this study, there is partial flexibility in terms of equipment (i.e. for each operation, there is an equipment subset able to perform the analysis) and, depending on the product, only some of the analysts are able to assist the analysis. Furthermore, the equipments and analysts were considered to be free for all jobs, having no dedicated lines.

Quality control laboratories are structures that support production. Hence, the main objective when considering this facilities workflow is to process the samples sent from production as fast as possible, Maslaton (2012).

The quality control laboratory scheduling is optimally designed by solving the following problem:

- Given:
  (1) a set of samples that require a set of analytical procedures to be performed in a given order;
  (2) available equipment and corresponding suitability for each sample analysis procedure;
  (3) each analytical procedure has a specific, fixed processing time, dependent on which equipment the analysis will be done;
  (4) available analysts and corresponding suitability for each sample analysis procedure;
  (5) each analytical procedure requires an analyst at a given time during a specific duration;
  (6) an analytical procedure may require the analyst more than once during its processing.
- Determine:
  (1) an optimal allocation to an equipment;
  (2) an optimal allocation to an analyst;
  (3) an optimal sequence of samples and analytical procedures for an equipment;
  (4) an optimal sequence of samples and analytical procedures for an analyst.
- So as to:
  (1) minimize maximum completion time.

In the following section the mathematical formulation to solve the problem stated above is described.

3. MATHEMATICAL FORMULATION

In this section the proposed MILP is presented for the dual resource constraint scheduling problem, and adapted to the case-study presented. Both modelling assumption, constraints and objective function are thoroughly described.

3.1 Time modelling

Scheduling formulations are very dependent on the time modelling strategy. Time can be modelled in a discrete or a continuous way, Floudas and Lin (2005).

The time discretisation approach involves splitting the time horizon in a number of equal duration time intervals. Events such as the start/end of a task can only take place at those time intervals. It is usually necessary to use a short length for the time units to achieve a good time approximation. However, depending on the relation between time unit duration and time horizon, this may imply very large models.

In continuous-time approaches, the durations of intervals are treated as variables. An interval starts or ends at an event and its duration is a continuous variable. The time grid can be common to all resources or different time grids for each resource can be used. In this work a continuous-time approach using the two different time grid strategies was used. A common time grid was employed for the worker resources and resource differing time grids were adopted for machines. This way, not only was time modelled in a very accurate way, but a much smaller number of events need to be processed for the machines. The same option was not taken for workers because these are required to be present in various distinct times during the processing of operations and can, in the meantime, engage in other operations.

3.2 Objective function

The objective in this case study was to perform the best scheduling that enables the optimum completion time for the tests in the laboratory. Hence, the objective is to minimise makespan, equation (1). The nomenclature is presented in Annex A.

$$min \quad C_{max} \tag{1}$$

3.3 Constraints

Test precedence constraints Precedence constraints enforce that a test can only start after the completion of the previous test, (2). Since a test processing time is considered deterministic and only depends on the equipment processing it, the completion time of the test is given by its starting time plus the processing time on the chosen equipment.

$$t(s,o) + \sum_{k,l} \mu(s,o,k) X(s,o,k,l) \leq t(s,o+1), \quad \forall s \in S, o \in O_s, o \neq |O_s|, k \in K_{O_s} \tag{2}$$

Equipment precedence constraints In an equipment, a test can only start after the previous test was completed. Similarly to (2), the time a test can start on an equipment has to be at least the time a previous test performed by the same equipment was completed, (3).

$$T(k,l) + \sum_{s,o} \mu(s,o,k) X(s,o,k,l) \leq T(k,l+1), \quad \forall s \in S, o \in O_s, k \in K_{O_s}, l \neq l_{last} \tag{3}$$

Analysts start and end time constraints The analyst allocation variable, $Xa(s,o,a,\theta,n)$, is dependent on $\theta = \{1,..,\tau(s,o)\}$, where $\tau(s,o)$ is the maximum number of times the analyst's presence is requested by that test. The indices θ indicates the sequence request within a test, ex. if the first time an analyst is requested for that test is at the time of setup, then the θ for setup will be 1.

For the definition of the analyst's starting and ending time for processing of a sample test, five constraints are enforced.

Equation (4) sets the starting time of an analyst to be at least the time the tests requests the analyst's presence. Equation (5) enforces that, if a test is assigned to an analyst at a time point n, the analyst needs to be there at exactly that time. Otherwise the analyst's time is not bounded.

$$T_s(s,a,n) \geq t(s,o) - M(1 - X_a(s,o,a,\theta,n)) + \sum_{\theta}^{\tau(s,o)} A^s(s,o,a,\theta) X_a(s,o,a,\theta,n), \quad \forall s \in S, o \in O_s, a \in A_{O_s} \tag{4}$$

$$T_s(s,a,n) \leq t(s,o) + M(1 - X_a(s,o,a,\theta,n)) + \sum_{\theta}^{\tau(s,o)} A^s(s,o,a,\theta) X_a(s,o,a,\theta,n), \quad \forall s \in S, o \in O_s, a \in A_{O_s} \tag{5}$$

Equation (6) defines the ending time of an analyst intervention as the starting time of the intervention, $Ts(s,a,n)$, plus its duration. Equation (7) is a bounding equation that limits the ending time of an intervention of an analyst in a test to the end of the test processing time.

$$T_e(s,a,n) = T_s(s,a,n) + \sum_{\theta}^{\tau(s,o)} A^d(s,o,a,\theta) X_a(s,o,a,\theta,n), \quad \forall s \in S, o \in O_s, a \in A_{O_s} \tag{6}$$

$$T_e(s,a,n) \leq t(s,o) + M(1 - X_a(s,o,a,\theta,n)) + \sum_{k,l} \mu(s,o,k) X(s,o,k,l), \quad \forall s \in S, o \in O_s, a \in A_{O_s}, k \in K_{O_s} \tag{7}$$

Finally, equation (8) sets the precedence constraints for each analyst to ensure no overlap of tasks.

$$T_s(s,a,n+1) \geq T_e(j,a,n), \quad \forall s,j \in S, o \in O_s, a \in A_{O_s}, n \neq n_{last} \tag{8}$$

Allocation constraints Allocation constraints ensure that the real system capacity is respected and demand is met. Both the equipment allocation variable and the analyst allocation variable are bounded by this constraints.

Equations (9) and (10) are equipment related. Equation (9) ensures a sample test is always performed on an equipment suitable to it. Equation (10) limits the number of tests being allocated to an equipment to be at most 1 at a time.

$$\sum_{k,l} X(s,o,k,l) = 1, \forall s \in S, o \in O_s, k \in K_{O_s} \tag{9}$$

$$\sum_{s,o} X(s,o,k,l) \leq 1, \forall s \in S, o \in O_s, k \in K_{O_s} \tag{10}$$

Equations (11) and (12) are equipment related. Equation (11) ensures a sample test is always allocated to an analyst each time the analyst's presence is requested. Equation (12) states that an analyst can only be allocated to one test at a time.

$$\sum_{a,n} X_a(s,o,a,\theta,n) = 1, \quad \forall s \in S, o \in O_s, a \in A_{O_s}, \theta \leq \tau(s,o) \tag{11}$$

$$\sum_{s,o,\theta} X_a(s,o,a,\theta,n) \leq 1, \quad \forall s \in S, o \in O_s, a \in A_{O_s}, \theta \leq \tau(s,o) \tag{12}$$

Linking constraints There is the need to link the starting time of a test in an equipment with the equipment starting time, so equations (13) and (14) are defined.

$$T(k,l) \leq t(s,o) + M(1 - X(s,o,k,l)), \quad \forall s \in S, o \in O_s, k \in K_{O_s} \tag{13}$$

$$T(k,l) + M(1 - X(s,o,k,l)) \geq t(s,o), \quad \forall s \in S, o \in O_s, k \in K_{O_s} \tag{14}$$

Makespan definition Makespan is the completion time of the last test to be completed. This constraint cannot be formulated using a *max* operator since it would be non-linear, and so it has to be enforced using multiple constraints, equation (15).

$$C_{max} \geq t(s,o) + \sum_{k,l} \mu(s,o,k) X(s,o,k,l), \quad \forall s \in S, o \in O_s, k \in K_{O_s} \tag{15}$$

4. RESULTS AND DISCUSSION

From the presented mathematical formulation, a mixed integer linear programming model was implemented. It was implemented in the General Algebraic Modelling System (GAMS) and solved using an Intel core i7 2.50 GHz processor and 16GB RAM. The commercial solver used was CPLEX. Due to the operational nature of the scheduling problem, the stopping criterion chosen was a maximum computational time of one hour.

To develop the scenarios, three types of jobs were created. These were based on the product types in Ruiz-Torres et al. (2012), and are presented in Table 1. Job type A has three associated operations that have precedence between them (e.g. operation 2 can only start after the completion of operation 1), job type B has two operations with precedence constrains as well, and job type C has only one associated operation.

Each job type requires an analyst at different times throughout its processing. The starting time and duration of these analyst tasks is presented in Table 2. For job type C and operations 2 and 3 of job type A analysts are only required for setup and teardown. In both operations of job type B and operation 1 of job type A, analysts are needed during a specific part of the processing as well.

Table 1. Job types and associated characteristics.

Job type	Operation	Processing time
A	1	3
	2	2
	3	1
B	1	5
	2	4
C	1	1

Table 2. Job analyst requirements in terms of start time and duration of the activity.

Job type	Oper.	Setup		Intermediate		Teardown	
		Start	End	Start	End	Start	End
A	1	0	0.25	1	2	2.75	3
	2	0	0.25	-	-	0.75	1
	3	0	0.25	-	-	0.75	1
B	1	0	0.25	2	3	4	5
	2	0	0.25	1	2	3	4
C	1	0	0.25	-	-	0.75	1

Testing scenarios were characterized by number of jobs A, B and C. Two types of problems were tested, small and medium-large instances. The former was intended as an illustrative example. The latter are instances created to mimic a daily workload in a real QC laboratory, based on the data reported by Costigliola et al. (2017). The total number of jobs considered for the small instance was 3, one of each type. For medium-large instances the number of jobs varied between 30 and 40 jobs. However, since this is a flexible job shop problem and jobs are comprised of more than one operation, it is also important to consider the number of operations in each instance. Therefore, the small instance has 6 operations, while the medium-large instance has between 35 and 52 operations.

4.1 Small instance scenario

The small instance scenario was created as a simple illustrative example. Therefore, three jobs, one of each type, and three machines were used. The association between machines and job types can be seen on Table 3. Furthermore, these instances because of their size had only one analyst, as to be sure the problem was constrained by human resources.

Table 3. Resource suitability for the small instance

No. jobs	Job type	Suitable analysts	Suitable machines
< 3	A	a1	m1 or m2
	B	a1	m1, m2 or m3
	C	a1	m1 or m3

The results for the small instance are shown in Table 4. The model was solved to optimality in 34.52 seconds. In Figure 1 and Figure 2, an optimal solution is shown, having a makespan of 11. It is possible to observe in the figures both the analyst's presence in intermediate time points during the operations' processing and the analyst shifting between operations.

The model statistics are presented in Table 5. With higher complexity problems, not just in terms of number of operations but also in terms of number of resources,

Table 4. Results for small instances

No. Jobs A	B	C	Total no. operations	C_{max}	Gap	CPU time (s)
1	1	1	6	11	0	34.52

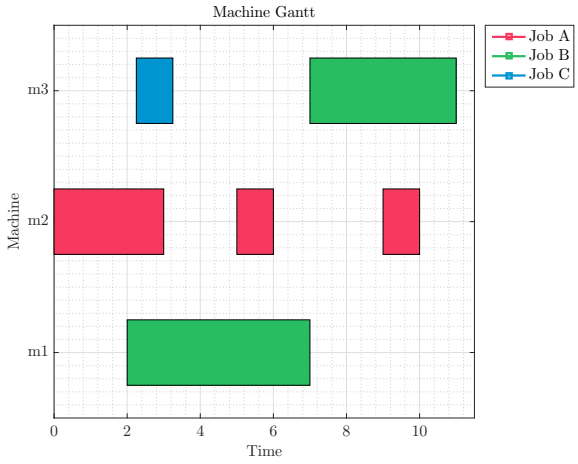


Fig. 1. Equipment Gantt chart for the small instance scenario with one job of each type. An optimal solution was obtained ($C_{max} = 11$).

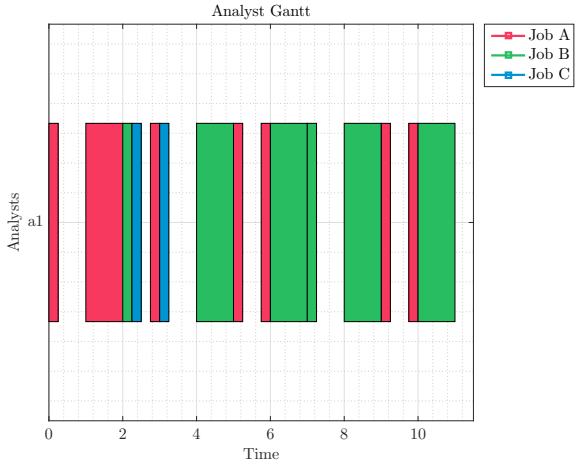


Fig. 2. Analyst Gantt chart for the small instance scenario. An optimal solution was obtained ($C_{max} = 11$).

the model statistics increase significantly. This is due to the fact that there is at least one variable per resource and there are variables associated with the resource time points. The number of time points is a tuning parameter, if it is too large it significantly increases the model size, and if it is too small it may compromise the optimality of the solution.

Table 5. Model statistics for the small instance

Scenario (N_A,N_B,N_C)	(1,1,1)
Block of equations	15
Single equations	1231
Blocks of variables	8
Single variables	830
Non zeros elements	5811
Discrete variables	624
Computational time (s)	34.52

4.2 Medium-large instances

Medium-large instances have between 30 and 40 jobs and were developed to mimic reality. This range of jobs yield a number of operations between 35 and 52, which is in accordance with the values stated by Costigliola et al. (2017). In this study it is considered that 85% of the samples have only one test to perform (similarly to sample type C), around 10% have two tests (resembling sample type B), and around 5% have three tests to perform.

As resources, fifteen machines and five analysts were considered. Because the amount of samples of type C is significantly larger than the amount of samples of the other types, there are more machines able to analyse type C samples. Furthermore, there were more analysts considered able to analyse samples of type C than of type A and B, as presented in Table 6. Compared to the previous instance, the medium-large instances consist of a significantly complexity increase not just because of the higher amount of resources and jobs but also because of the higher flexibility present, namely in job type C.

Table 6. Resource suitability for medium-large instances

No. jobs	Job type	Suitable analysts	Suitable machines
≥ 30	A	a1	m15
	B	a2	m12 to m15
	C	a3 to a5	m1 to m12

The results are shown in Table 7. The model was tested for a total of 30 and 40 jobs. It is possible to see that both instances were not solved optimally for a maximum computational time of 1h. The gap presented in Table 7 corresponds to the relative gap between the best known integer solution and the lower bound calculated through the linear relaxation. It is enough that the linear relaxation is not very strong for the computed gap to be very different from the actual distance between the best known integer solution and the optimal.

Table 7. Results for medium-large instances

No. Jobs A	B	C	Total no. operations	C_{max}	Gap	CPU time (h)
1	3	26	35	27	66.67%	1
2	4	34	52	36	75.0%	1

In Table 8 the statistics for the medium-large instances are presented. By comparing the two instances it is possible to see the large increase in equations and variables, evidencing the whole problem complexity.

Table 8. Model statistics for the medium-large instances

Scenario	(1,3,26)	(2,4,34)
Block of equations	15	15
Single equations	118698	189586
Blocks of variables	8	8
Single variables	31747	42710
Non zeros elements	893219	1240706
Discrete variables	22410	30360
Computational time (h)	1	1

5. CONCLUSIONS

This paper develops a mathematical model for scheduling problem in quality control laboratories. The problem is formulated as a dual resource constrained flexible job shop problem, in which workers are not only required for setup and teardown operations, but also for intermediate points throughout the processing of a task. From the proposed mathematical formulation, a MILP was developed and implemented. The model was then tested on small and medium-large instances. Results show that the model was able to achieve optimality for the small instance. For the two medium-large instances the model was able to find a solution, but optimality was not guaranteed. The problems with finding the optimal solution were related to the instances increase in complexity.

As future work, research should focus on exploring additional solution methods for the quality control scheduling problem. Constraint programming and hybrid approaches, the latter combining constraint and integer programming, may be more effective in such a large model. Furthermore, due to the highly combinatorial nature of this problem, we believe heuristics or metaheuristics will also be able to achieve good solutions in a shorter computational time than MILP. Further testing may also be of interest to ascertain the model complexity and how it is affected by the test instance structure.

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Appendix A. NOMENCLATURE

Sets
- S is the set of samples
- O_s is the set of tests from sample s
- K_{O_s} is the set of equipment suitable for processing the test O_s
- A_{O_s} is the set of analysts suitable for processing the test O_s

Indices
- s, j = samples
- o = test
- k = equipment
- a = analyst
- l = event point for equipment
- n = event point for analysts
- θ = relative time for tests

Parameters
- $\mu(s, o, k)$ = processing time of the test o of sample s in equipment k
- $A^s(s, o, a, \theta)$ = time after the start of the processing of test o of sample s that intervention θ by analyst a is requested
- $A^d(s, o, a, \theta)$ = duration of intervention θ by analyst a requested by test o of sample s
- M = big number
- $\tau(s, o)$ = maximum number of analyst intervention needed for test o of sample s

Variables
- $X(s, o, k, l) = 1$ if test o of sample s starts in equipment k at event point l; 0 otherwise
- $t(s, o)$ = starting time of test o from sample s
- $T(k, l)$ = starting time of time point l of equipment k
- $X_a(s, o, a, \theta, n) = 1$ if intervention θ of test o from sample s is assisted by analyst a at time point n; 0 otherwise
- $T_s(s, a, n)$ = starting time of sample s on time point n of analyst a
- $T_e(s, a, n)$ = end time of sample s on time point n of analyst a
- C_{max} = makespan